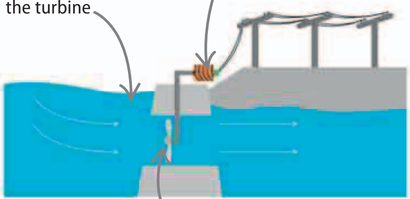
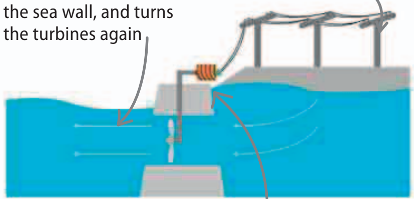
## Tidal power

The movements of the sea can be converted into usable power in a number of ways. In one type of tidal power system, both the incoming and outgoing tides produce electricity by turning turbines.

▽ **Inward tide**
The incoming tide passes through a gap in a sea wall and turns a turbine mounted in the gap.




▽ **Outward tide**
When the tide goes out, the water flows back through the gap, turning the turbines again, generating more electricity.




REAL WORLD

### Energy from the waves

The Pelamis wave energy converter draws power from sea waves. The converter is made of floating sections that are hinged together. As they bend with the waves, the "hinges" force fluid along pipes, and the pressure of the fluid is used to rotate turbines and generate electricity.

## Geothermal energy

The interior of Earth is hotter than the surface—on average, the temperature increases by an interval of around 30°C (50°F) for every kilometer (0.6 miles) of depth. The difference between thesurface and underground temperatures can be exploited to generate electricity, or simply to heat water for domestic use.

▷ **Electricity from underground**
In this geothermal system, water is pumped underground and pushes groundwater up to the surface. This water contains mineral salts and is referred to as geothermal brine. The brine is hot enough to boil and the high-pressure steam produced is used to turn turbines in a power station, generating electricity. The power station then distributes this electricity to the power grid, so it can be used in homes and buildings.